SUR5365 - Digital Mapping

1. OVERVIEW

This course covers theoretical concepts and practical aspects for mapping and analyzing digital spatial data. It is comprised of lectures and computer labs and uses various software packages to visualize, integrate, and analyze spatial data. Various online resources are presented to obtain free geodata and to present data in online maps.

- Fall semester, 3 credits
- 100% online, synchronous and asynchronous component
- http://elearning.ufl.edu/

Course Prerequisites:
No formal course prerequisites. Prior experience with ArcGIS Pro software, Microsoft Excel, and scripting/programming in any programming language (e.g., Python, R, Java) is recommended. A basic understanding of analytic geometry, trigonometry, and statistics is required.

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- Please use the Canvas message/Inbox feature for fastest response
- Virtual office hours by appointment

Lectures:
Fridays: 11:45 am - 2:30 pm (per. 5-7), via Zoom; links to lecture recordings and Q&A sessions are provided on the course website

Office hours (Q&A):
Thursdays from 5-6 pm in Zoom; attendance is optional

Recommended reading materials:
- No course book is required
- References to books, book chapters, and online resources will be given during the lecture periods
- Short instructional videos closely related to the lecture content can be found at the Geomatics @ FLREC YouTube channel

Software requirements:
- ArcGIS Pro 3.5.x and Microsoft Excel will be used for many topics. If unfamiliar with ArcGIS Pro, it is recommended to run through practice tutorials beforehand. ArcGIS Pro download and installation instructions are provided on the course website under the Week 1 module.
- Additional free software packages used (e.g., HTML editors) will be introduced in the different course modules.

2. LEARNING OUTCOMES

The course objective is to provide students with the following competencies at the completion of the course:
1. Use map projections in combination with geodetic datums for mapping
2. Apply critical thinking skills in GIS including the interpretation of computational results
3. Read and extract information from digital topographic maps

4. Visualize spatial data through Web mapping and scripting
5. Use AI tools for spatial data analysis and visualization

3. COURSE LOGISTICS

- Throughout the semester, students will be given 9 home assignments, 5 quizzes, and 3 online discussions. For each assignment/quiz/discussion, a due date and time is given, which is the beginning of the next class (11:45am Friday).
- Assignments are graded based on timeliness, correctness of computations, and interpretation of numerical results with written feedback on student creativity and technical ability provided by the instructor; quizzes are auto-graded based on correctness of multiple choice questions with correct answers shown after completion; and discussion items are graded within a week based on creativity, completeness, technical correctness and the number of comments provided to peers.
- There is a 1-week turnaround for assignment and discussion grading. Quizzes are autograded instantaneously in Canvas.
- This course is a distance education course taught as live lectures using the virtual classroom software Zoom. Lecture materials can be downloaded from weekly modules on the Canvas website.

The Canvas system should be used as the platform for written communication between students and the instructors. Questions and suggestions to the whole class can also be posted under the Discussions tab. Any short-term changes concerning lectures or other course components will be announced through Canvas. Feel free to call the instructors with any questions.

Technology Requirements:

- A computer or mobile device with high-speed internet connection
- ArcGIS Pro runs only on Microsoft operating systems. If students use a Mac computer or other operating systems, they are encouraged to use ArcGIS Pro in UF Apps (https://info.apps.ufl.edu/).
- A headset and/or microphone and speakers to participate in live sessions
- For Zoom: A supported web browser on a supported operating system (Windows, Mac OS, Linux); and minimum bandwidth. More details can be found here.

Using Zoom:

Live lectures and office hour meetings (per individual student requests) will be conducted with the Zoom conferencing software. Sessions can be joined by clicking a link posted by the instructor on Canvas.

Grades:

Item	***Percentage***
Home assignments (9 assignments @ 8% each)	72%
Quizzes (5 quizzes @ 3.5% each)	17.5%
Discussions (3 discussions @ 3.5% each)	10.5%
Total	**100%**

Grading scale:

Grade	*Percentage*	*Grade*	*Percentage*
A	92.0-100.0	C+	78.0-79.9
A-	90.0-91.9	C	72.0-77.9
B+	88.0-89.9	C-	70.0-71.9
B	82.0-87.9	D	60.0-69.9
B-	80.0-81.9	E	0-59.9

For information on current UF policies for assigning grade points, see https://catalog.ufl.edu/UGRD/academic-regulations/grades-grading-policies/

4. COURSE CONTENT

Week	Topic	Assignment (due)
Week 1, Aug 22	Introduction Coordinate systems and geodetic datums	D1, Q1 (Aug 29)
Week 2, Aug 29	Map projections: Introduction Cylindrical projections	Q2, H1 (Sep 5)
Week 3, Sep 5	Cylindrical map projections (cont.) Conic projections	Q3 (Sep 12)
Week 4, Sep 12	Azimuthal projections Topographic maps: introduction	H2, D2 (Sep 19)
Week 5, Sep 19	Topographic maps (cont.): coordinates and elevations	H3, Q4 (Sep 26)
Week 6, Sep 26	Introduction to HTML	H4 (Oct 3)
Week 7, Oct 3	Introduction to JavaScript	H5 (Oct 10)
Week 8, Oct 10	Web mapping; JavaScript libraries	H6 (Oct 24)
Week 9, Oct 17	Homecoming - No classes	
Week 10, Oct 24	ArcGIS Online	H7 (Nov 7)
Week 11, Oct 31	Q&A session (attendance optional, no lecture)	
Week 12, Nov 7	GeoAI tools	H8 (Nov 14)
Week 13, Nov 14	Coordinate and datum transformations	H9, Q5 (Nov 21)
Week 14, Nov 21	Google Earth	D3 (Dec 3)
Week 15, Nov 28	Thanksgiving - No classes	

D: Discussion, Q: Quiz, H: Home assignment

5. ACADEMIC POLICIES and RESOURCES

Late submissions and make-up requests:

It is the responsibility of the student to access online lectures, readings, quizzes, and exams and to maintain satisfactory progress in the course.

- A 10% penalty per day will be applied to late assignments. A late submission on the due date also results in a 10% deduction.
- Assignments will not be accepted if handed in more than seven days after the due date.
- Quizzes cannot be taken past the deadline.
- Online discussions cannot be completed past the deadline.
- Exceptions to the late policy are only allowed per university policy.
- There will be no extra credit assignments.

Computer or other hardware failures, except failure of the UF canvas system, will not excuse students for missing assignments. Any late submissions due to technical issues MUST be accompanied by the ticket number received from the Helpdesk when the problem was reported to them. The ticket number will document the time and date of the problem. You MUST e-mail your instructor within 24 hours of the technical difficulty if you wish to request consideration. For computer, software compatibility, or access problems call the HELP DESK phone number—352-392-HELP = 352- 392-4357 (option 2).

Requirements for class attendance and make-up exams, assignments and other work are consistent with university policies that can be found at: https://catalog.ufl.edu/UGRD/academic-regulations/attendance-policies/

Netiquette: Communication Courtesy:
All members of the class are expected to follow rules of common courtesy in all email messages, threaded discussions and chats, as laid out in the UF Netiquette Guide for Online Courses. Failure to do so may result in loss of participation points and/or referral to the Dean of Students' Office.

For other academic policies regarding semester evaluation process, the University's Honesty Policy, in-class recording, and University Policy on accommodating students with disabilities as well as academic resources and campus health and wellness resources see https://syllabus.ufl.edu/syllabus-policy/uf-syllabus-policy-links/